

MathLive 数式入力 安定性テスト (50式)

各数式をMathLive入力欄に1つずつ手入力してテストする。
 入力順序が重要：関数名の連続入力で既存構造が壊れないか確認。

$$\sum_{k=0}^{\infty} \frac{(-1)^k}{(2k+1)!} \left(\sum_{n=1}^N \sin(n\theta) \cos(n\phi) \right)^{2k+1} = \sin \left(\sum_{n=1}^N \sin(n\theta) \cos(n\phi) \right)$$

$$\prod_{p \text{ prime}} (1-p^{-s})^{-1} = \sum_{n=1}^{\infty} \frac{\mu(n)^2}{\phi(n)} \cdot \frac{\prod_{j=1}^{\omega(n)} \sin(\pi p_j^{-s})}{\sum_{d|n} \Lambda(d) \log^2 d}$$

$$\iiint_{\mathbb{R}^3} \frac{\partial^3}{\partial x \partial y \partial z} \left[\frac{e^{-\alpha \sqrt{x^2+y^2+z^2}} \sin(\beta \sqrt{x^2+y^2+z^2})}{(x^2+y^2+z^2)^{3/2}} \right] dx dy dz$$

$$\det \begin{pmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} & \cdots & \frac{\partial f_1}{\partial x_n} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} & \cdots & \frac{\partial f_2}{\partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial f_n}{\partial x_1} & \frac{\partial f_n}{\partial x_2} & \cdots & \frac{\partial f_n}{\partial x_n} \end{pmatrix} = \prod_{i=1}^n \lambda_i$$

$$\lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=0}^{N-1} \exp \left(2\pi i \sum_{j=1}^k \alpha_j \lfloor n\beta_j \rfloor \right) = \prod_{j=1}^k \lim_{M \rightarrow \infty} \frac{1}{M} \sum_{m=0}^{M-1} e^{2\pi i \alpha_j \lfloor m\beta_j \rfloor}$$

$$\frac{1}{1+\frac{e^{-2\pi}}{1+\frac{e^{-4\pi}}{1+\frac{e^{-6\pi}}{1+\frac{e^{-8\pi}}{\ddots}}}}}}=\left(\sqrt{\frac{5+\sqrt{5}}{2}}-\frac{\sqrt{5}+1}{2}\right)e^{2\pi/5}$$

$$R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}-4R_{\mu\nu}R^{\mu\nu}+R^2=\frac{1}{(4\pi)^{n/2}}\frac{\Gamma(n/2-1)}{\Gamma(n)}\int_{\mathcal{M}}\sqrt{|g|}\,\epsilon^{\mu_1\cdots\mu_n}\epsilon^{\nu_1\cdots\nu_n}\prod_{k=1}^{n/2}R_{\mu_{2k-1}\mu_{2k}\nu_{2k-1}\nu_{2k}}d^nx$$

$$\hat{H}\Psi=\left[-\sum_{i=1}^N\frac{\hbar^2}{2m_i}\nabla_i^2+\sum_{i<j}\frac{e^2}{4\pi\epsilon_0|\mathbf{r}_i-\mathbf{r}_j|}-\sum_{i=1}^N\sum_{A=1}^M\frac{Z_Ae^2}{4\pi\epsilon_0|\mathbf{r}_i-\mathbf{R}_A|}\right]\Psi=E\Psi$$

$$\mathcal{Z}[\beta]=\int \mathcal{D}[\phi] \, \exp\left(-\frac{1}{\hbar}\int_0^{\beta\hbar} d\tau \int d^3\mathbf{x} \left[\frac{1}{2}(\partial_\mu\phi)^2+\frac{m^2}{2}\phi^2+\frac{\lambda}{4!}\phi^4+\sum_{n=1}^{\infty}\frac{g_n}{n!}\phi^n\right]\right)$$

$$s_{\lambda/\mu}(x_1,\ldots,x_n)=\sum_{T\in\mathrm{SSYT}(\lambda/\mu,n)}\prod_{i=1}^nx_i^{|T^{-1}(i)|}=\frac{\det(h_{\lambda_i-\mu_j-i+j}(x_1,\ldots,x_n))_{1\leq i,j\leq\ell}}{\prod_{1\leq i<j\leq n}(x_i-x_j)}$$

$$\left(\prod_{k=1}^K\sum_{n_k=0}^{\infty}\frac{z_k^{n_k}}{n_k!}\right)\exp\left(-\sum_{k=1}^Kz_k\right)=\sum_{\mathbf{n}\in\mathbb{Z}_{\geq 0}^K}\left(\prod_{k=1}^K\frac{z_k^{n_k}e^{-z_k}}{n_k!}\right)$$

$$\mathrm{tr}\big[\gamma^5\gamma^{\mu_1}\gamma^{\mu_2}\cdots\gamma^{\mu_{2n}}\big]=(-2i)^n\sum_{\sigma\in S_{2n}}\mathrm{sgn}(\sigma)\prod_{k=1}^n\eta^{\mu_{\sigma(2k-1)}\mu_{\sigma(2k)}}\cdot\varepsilon^{\mu_1\mu_2\cdots\mu_{2n}}$$

$$\frac{\partial}{\partial t}\begin{pmatrix}\rho\\ \rho u\\ \rho v\\ \rho E\end{pmatrix}+\frac{\partial}{\partial x}\begin{pmatrix}\rho u\\ \rho u^2+p\\ \rho uv\\ (\rho E+p)u\end{pmatrix}+\frac{\partial}{\partial y}\begin{pmatrix}\rho v\\ \rho uv\\ \rho v^2+p\\ (\rho E+p)v\end{pmatrix}=\begin{pmatrix}0\\ \rho f_x\\ \rho f_y\\ \rho (uf_x+vf_y)+\nabla\cdot(k\nabla T)\end{pmatrix}$$

$$\Gamma_{\mu\nu}^\lambda=\frac{1}{2}g^{\lambda\sigma}\Big(\frac{\partial g_{\sigma\mu}}{\partial x^\nu}+\frac{\partial g_{\sigma\nu}}{\partial x^\mu}-\frac{\partial g_{\mu\nu}}{\partial x^\sigma}\Big),\quad R^\rho{}_{\sigma\mu\nu}=\partial_\mu\Gamma^\rho{}_{\nu\sigma}-\partial_\nu\Gamma^\rho{}_{\mu\sigma}+\Gamma^\rho{}_{\mu\lambda}\Gamma^\lambda{}_{\nu\sigma}-\Gamma^\rho{}_{\nu\lambda}\Gamma^\lambda{}_{\mu\sigma}$$

$$\int_0^\infty \frac{\sin^{2n+1}(x)}{x^{2n+1}}\,dx = \frac{(-1)^n\pi}{2^{2n+1}(2n)!}\sum_{k=0}^n (-1)^k\binom{2n+1}{n-k}(2k+1)^{2n}\operatorname{sgn}(2k+1)$$

$$\underbrace{\nabla\times(\nabla\times\mathbf{E})}_{\text{curl of curl}}=\underbrace{\nabla(\nabla\cdot\mathbf{E})}_{\text{grad of div}}-\underbrace{\nabla^2\mathbf{E}}_{\text{Laplacian}}=-\mu_0\varepsilon_0\frac{\partial^2\mathbf{E}}{\partial t^2}$$

$$\zeta(s)=\frac{1}{\Gamma(s)}\int_0^\infty\frac{t^{s-1}}{e^t-1}\,dt=2^s\pi^{s-1}\sin\Big(\frac{\pi s}{2}\Big)\,\Gamma(1-s)\zeta(1-s),\quad \Re(s)>1$$

$$\left\langle \Omega \left| T \left\{ \prod_{i=1}^n \phi(x_i) \right\} \right| \Omega \right\rangle = \sum_{\text{contractions pairs}(i,j)} \prod D_F(x_i-x_j) \cdot \sum_{\text{loops}} \frac{(-i\lambda)^V}{S} \prod_{l=1}^L \int \frac{d^4 k_l}{(2\pi)^4} \prod_I \frac{i}{k_I^2-m^2+i\varepsilon}$$

$$\widehat{\mathcal{F}}[f](\xi)=\int_{\mathbb{R}^n}f(\mathbf{x})\,e^{-2\pi i\langle \mathbf{x},\xi\rangle}\,d\mathbf{x},\quad f(\mathbf{x})=\int_{\mathbb{R}^n}\widehat{\mathcal{F}}[f](\xi)\,e^{2\pi i\langle \mathbf{x},\xi\rangle}\,d\xi$$

$$\sum_{n=0}^\infty \frac{H_n^{(r)}}{n^s} z^n = \frac{1}{\Gamma(s)} \int_0^\infty \frac{t^{s-1}}{e^t - z} \operatorname{Li}_r(ze^{-t}) \, dt, \quad H_n^{(r)} = \sum_{k=1}^n \frac{1}{k^r}$$

$$\left|\begin{array}{cccc} a_{11}-\lambda & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22}-\lambda & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn}-\lambda \end{array}\right| = \sum_{k=0}^n (-1)^k e_{n-k}(\lambda_1,\ldots,\lambda_n) \lambda^k$$

$$D_\mu F_a^{\mu\nu} + g f_{abc} A_b^\mu F_{\mu\,c}^{\,\,\nu} = J_a^\nu, \quad F_{\mu\nu}^a = \partial_\mu A_\nu^a - \partial_\nu A_\mu^a + g f^{abc} A_\mu^b A_\nu^c$$

$$\mathrm{Pf}(A)=\frac{1}{2^nn!}\sum_{\sigma\in S_{2n}}\mathrm{sgn}(\sigma)\prod_{i=1}^na_{\sigma(2i-1),\sigma(2i)},\quad \mathrm{Pf}(A)^2=\det(A)$$

$$\left[\hat{a}_{\mathbf{k}\lambda},\hat{a}_{\mathbf{k}'\lambda'}^\dagger\right]=(2\pi)^32\omega_{\mathbf{k}}\,\delta^{(3)}(\mathbf{k}-\mathbf{k}')\delta_{\lambda\lambda'},\quad \hat{A}^\mu(x)=\sum_{\lambda=1,2}\int\frac{d^3k}{(2\pi)^32\omega_{\mathbf{k}}}\left[\epsilon_\lambda^\mu(\mathbf{k})\hat{a}_{\mathbf{k}\lambda}e^{-ik\cdot x}+\bar{\epsilon}_\lambda^\mu(\mathbf{k})\hat{a}_{\mathbf{k}\lambda}^\dagger e^{ik\cdot x}\right]$$

$$K(\mathbf{x},\mathbf{x}')=\sigma_f^2\exp\left(-\frac{1}{2}\sum_{d=1}^D\frac{(x_d-x'_d)^2}{\ell_d^2}\right)+\sigma_n^2\delta(\mathbf{x}-\mathbf{x}'),\quad \mathbf{f}_*|\mathbf{X}_*,\mathbf{X},\mathbf{y}\sim\mathcal{N}(K_*^T(K+\sigma_n^2I)^{-1}\mathbf{y},\,K_{**}-K_*^T(K+\sigma_n^2I)^{-1}K_*)$$

$$\int_{-\infty}^{\infty}\left|\sum_{n=-N}^N c_n e^{2\pi i n \alpha x}\cdot \text{sinc}(\beta x-n)\right|^2\frac{dx}{\cosh(\pi x/\gamma)+\cos(\pi\delta/\gamma)}=\frac{2\gamma}{\sin(\pi\delta/\gamma)}\sum_{n=-N}^N |c_n|^2 e^{-2\pi|n|\alpha\delta}$$

$$\mathcal{L}_{\text{QCD}} = \sum_q \bar{\psi}_{q,i} (i\gamma^\mu (D_\mu)_{ij} - m_q \delta_{ij}) \psi_{q,j} - \frac{1}{4} G_{\mu\nu}^a G^{a\mu\nu} + \frac{\theta g^2}{32\pi^2} G_{\mu\nu}^a \tilde{G}^{a\mu\nu}$$

$$\left(\sqrt[3]{\frac{\sqrt{a^2+b^2}+\sqrt{(a^2+b^2)^2-4a^2b^2\sin^2\theta}}{2}}+\sqrt[3]{\frac{\sqrt{a^2+b^2}-\sqrt{(a^2+b^2)^2-4a^2b^2\sin^2\theta}}{2}}\right)^3$$

$$\sum_{\substack{p_1+p_2+\cdots+p_k=n\\ p_i\geq 1}}\prod_{j=1}^k\left(\frac{1}{p_j!}\sum_{m=0}^{p_j}\binom{p_j}{m}(-1)^{p_j-m}\left(\frac{m}{n}\right)^{p_j}\right)\cdot\frac{n!}{\prod_{j=1}^kp_j!}$$

$$\mathbf{x}_{k+1} = \mathbf{x}_k - \left[\sum_{i=1}^m \nabla^2 f_i(\mathbf{x}_k) + \lambda \mathbf{I} \right]^{-1} \left[\sum_{i=1}^m \nabla f_i(\mathbf{x}_k) + \lambda(\mathbf{x}_k - \mathbf{x}_0) \right]$$

$$\Phi(x,t)=\frac{1}{\sqrt{2\pi\sigma^2t}}\int_{-\infty}^{\infty}\Phi_0(\xi)\exp\bigg(-\frac{(x-\xi-\mu t)^2}{2\sigma^2t}\bigg)d\xi+\sum_{n=1}^{\infty}A_n\sin\bigg(\frac{n\pi x}{L}\bigg)e^{-\kappa n^2\pi^2t/L^2}$$

$$\mathcal{W}[f](a,b)=\frac{1}{\sqrt{|a|}}\int_{-\infty}^{\infty}f(t)\overline{\psi\left(\frac{t-b}{a}\right)}dt,\quad f(t)=\frac{1}{C_{\psi}}\int_0^{\infty}\int_{-\infty}^{\infty}\mathcal{W}[f](a,b)\frac{1}{\sqrt{|a|}}\psi\left(\frac{t-b}{a}\right)\frac{db\,da}{a^2}$$

$$\chi^2=\sum_{i=1}^N\left(\frac{y_i-\sum_{j=0}^M\beta_j\phi_j(x_i)}{\sigma_i}\right)^2+\lambda\int_a^b\left(\sum_{j=0}^M\beta_j\phi_j''(x)\right)^2dx$$

$$\left\langle \prod_{i=1}^n V_{\alpha_i}(z_i, \bar{z}_i) \right\rangle = \left| \prod_{i < j} (z_i - z_j)^{2\alpha_i \alpha_j / \alpha_0^2} \right|^2 \cdot \delta \left(\sum_{i=1}^n \alpha_i - 2\alpha_0 \right)$$

$$\mathrm{Tr}\left[\mathcal{P}\exp\left(ig\oint_C A_\mu^aT^a dx^\mu\right)\right]=\sum_{n=0}^\infty (ig)^n\oint_{0\leq s_1\leq\cdots\leq s_n\leq 1}\prod_{k=1}^n A_\mu^{a_k}(x(s_k))\dot{x}^{\mu k}(s_k)\mathrm{Tr}[T^{a_1}\cdots T^{a_n}]ds_1\cdots ds_n$$

$$\hat{\rho}(t)=\hat{\rho}(0)-\frac{i}{\hbar}\int_0^t[\hat{H}(t'),\hat{\rho}(t')]dt'-\frac{1}{\hbar^2}\int_0^t\int_0^{t'}[\hat{H}(t'),[\hat{H}(t''),\hat{\rho}(t'')]]dt''dt'+\sum_k\gamma_k\left(\hat{L}_k\hat{\rho}\hat{L}_k^\dagger-\frac{1}{2}\{\hat{L}_k^\dagger\hat{L}_k,\hat{\rho}\}\right)$$

$$Z(\beta,\mu)=\prod_{\mathbf{k},\sigma}\sum_{n=0}^1\exp\Big[-\beta\left(\frac{\hbar^2k^2}{2m}-\mu\right)n\Big]=\prod_{\mathbf{k},\sigma}\Big(1+e^{-\beta(\epsilon_{\mathbf{k}}-\mu)}\Big),\quad\langle n_{\mathbf{k}\sigma}\rangle=\frac{1}{e^{\beta(\epsilon_{\mathbf{k}}-\mu)}+1}$$

$$\nabla_\mu T^{\mu\nu} = \partial_\mu T^{\mu\nu} + \Gamma^\mu_{\mu\lambda} T^{\lambda\nu} + \Gamma^\nu_{\mu\lambda} T^{\mu\lambda}, \quad \nabla_\mu \nabla_\nu V^\rho - \nabla_\nu \nabla_\mu V^\rho = R^\rho_{\sigma\mu\nu} V^\sigma$$

$$P(\theta|\mathcal{D})=\frac{P(\mathcal{D}|\theta)P(\theta)}{\int P(\mathcal{D}|\theta')P(\theta')\,d\theta'}\propto\prod_{i=1}^N\frac{1}{\sqrt{2\pi\sigma^2}}\exp\bigg(-\frac{(y_i-f_\theta(x_i))^2}{2\sigma^2}\bigg)\cdot\frac{1}{\sqrt{2\pi\tau^2}}\exp\bigg(-\frac{\|\theta\|^2}{2\tau^2}\bigg)$$

$$\frac{d}{dt}\begin{pmatrix} S \\ I \\ R \end{pmatrix} = \begin{pmatrix} -\beta SI/N \\ \beta SI/N - \gamma I \\ \gamma I \end{pmatrix} + \sqrt{\begin{pmatrix} \beta SI/N & -\beta SI/N & 0 \\ -\beta SI/N & \beta SI/N + \gamma I & -\gamma I \\ 0 & -\gamma I & \gamma I \end{pmatrix}} \begin{pmatrix} dW_1 \\ dW_2 \\ dW_3 \end{pmatrix}$$

$$\mathrm{KL}(p\|q)=\int_{\mathbb{R}^d}p(\mathbf{x})\log\frac{p(\mathbf{x})}{q(\mathbf{x})}d\mathbf{x}=\frac{1}{2}\left[\mathrm{tr}(\Sigma_q^{-1}\Sigma_p)+(\mu_q-\mu_p)^T\Sigma_q^{-1}(\mu_q-\mu_p)-d+\log\frac{|\Sigma_q|}{|\Sigma_p|}\right]$$

$$e^{i\pi\tau n^2+2\pi i n z}:\quad \vartheta_3(z|\tau)=\sum_{n=-\infty}^{\infty}e^{i\pi\tau n^2+2\pi i n z}=\prod_{m=1}^{\infty}(1-q^{2m})(1+q^{2m-1}e^{2\pi i z})(1+q^{2m-1}e^{-2\pi i z}),\quad q=e^{i\pi\tau}$$

$$\iint_S\left(\frac{\partial Q}{\partial x}-\frac{\partial P}{\partial y}\right)dA=\oint_{\partial S}P\,dx+Q\,dy,\quad\iiint_V\left(\frac{\partial P}{\partial x}+\frac{\partial Q}{\partial y}+\frac{\partial R}{\partial z}\right)dV=\oint\!\!\!\oint_{\partial V}P\,dy\,dz+Q\,dz\,dx+R\,dx\,dy$$

$$\mathcal{L}_{\rm Higgs}=(D_\mu\Phi)^\dagger(D^\mu\Phi)+\mu^2\Phi^\dagger\Phi-\lambda(\Phi^\dagger\Phi)^2,\quad\Phi=\frac{1}{\sqrt{2}}\begin{pmatrix}0\\v+h\end{pmatrix},\quad m_W=\frac{gv}{2},\quad m_Z=\frac{v\sqrt{g^2+g'^2}}{2}$$

$$f(z)=\sum_{n=-\infty}^{\infty}a_n(z-z_0)^n,\quad a_n=\frac{1}{2\pi i}\oint_C\frac{f(\zeta)}{(\zeta-z_0)^{n+1}}d\zeta,\quad \text{Res}_{z=z_0}f=a_{-1}=\lim_{z\rightarrow z_0}\frac{1}{(m-1)!}\frac{d^{m-1}}{dz^{m-1}}[(z-z_0)^mf(z)]$$

$$\Gamma(z)\Gamma(1-z)=\frac{\pi}{\sin(\pi z)},\quad \psi^{(n)}(z)=(-1)^{n+1}n!\sum_{k=0}^{\infty}\frac{1}{(z+k)^{n+1}},\quad B(x,y)=\frac{\Gamma(x)\Gamma(y)}{\Gamma(x+y)}=\int_0^{\pi/2}2\sin^{2x-1}\theta\cos^{2y-1}\theta\,d\theta$$

$$\hat{U}(t,t_0)=\mathcal{T}\exp\left(-\frac{i}{\hbar}\int_{t_0}^t\hat{H}_I(t')\,dt'\right)=\sum_{n=0}^{\infty}\frac{(-i/\hbar)^n}{n!}\int_{t_0}^t\cdots\int_{t_0}^t\mathcal{T}\left[\hat{H}_I(t_1)\cdots\hat{H}_I(t_n)\right]\,dt_1\cdots dt_n$$

$$\frac{\partial p}{\partial t} + \nabla \cdot (p \mathbf{v}) - \nabla \cdot \left(D \nabla p + \frac{p}{k_B T} \nabla U \right) = \sum_i \delta(t - t_i) \left[\prod_j \int K(\mathbf{x}, \mathbf{x}'_j | \theta_j) p(\mathbf{x}'_j, t_i^-) d\mathbf{x}'_j - p(\mathbf{x}, t_i^-) \right]$$

$$\int_0^1 \cdots \int_0^1 \left| \det \begin{pmatrix} K(x_1, y_1) & \cdots & K(x_1, y_n) \\ \vdots & \ddots & \vdots \\ K(x_n, y_1) & \cdots & K(x_n, y_n) \end{pmatrix} \right|^2 dx_1 \cdots dx_n dy_1 \cdots dy_n = n! \sum_{\sigma \in S_n} \prod_{k=1}^n \left(\int_0^1 |K(x, x)|^2 dx \right)^{c_k(\sigma)}$$

$$\mathcal{A}_n^{\text{tree}} = g^{n-2} \sum_{\sigma \in S_n/Z_n} \frac{\text{Tr}(T^{a_{\sigma(1)}} \cdots T^{a_{\sigma(n)}})}{\langle \sigma(1) \sigma(2) \rangle \langle \sigma(2) \sigma(3) \rangle \cdots \langle \sigma(n) \sigma(1) \rangle} \cdot \prod_{i < j} \left[\delta^{(8)} \left(\sum_{k=1}^n \lambda_k^\alpha \tilde{\lambda}_k^{\dot{\alpha}} \eta_k^A \right) \right]^{1/(n-2)}$$